Report

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Old Tulu Verb Morphology:
A Sketch

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Abstract: This work describes the verb morphology of Old Tulu, the language of the recently discovered epics that can be dated back to at least the 16th century. The following is an adapted version of an essay I wrote for a linguistics class during my freshman year of college. Much of this needs to be edited, and, as such, this will serve as a placeholder until I write an updated grammar of the language.

1 | INTRODUCTION

Tulu is a South Dravidian language of around three million speakers, used commonly in the South Canara and Kāsargōḍ districts of the Indian states of Karnāṭaka and Kēraḷa, respectively. The term "Tulunāḍṷ" refers to the region bounded by the Suvarṇa and Candragiri rivers in the north and south, and the Western Ghāṭs and the Arabian Sea in the west and east, where Tulu has traditionally been spoken. However, Tulunāḍṷ is a linguistically diverse region, home to speakers of Marāṭhī, Kannaḍa, Malayāḷam, Koṅkaṇi, and Koraga, resulting in strong (and sometimes clearly visible) influences, not just by those languages on Tulu, but also by Tulu on the other five.

Tulunāḍṷ has historically been divided along the Nētrāvati river into a north and south region, each of which has developed its own dialects.[1] In the same way that Sanskrit contact developed a distinct "Brahmin Tamiḻ" dialect, both regions developed their own "Brahmin Tulu" dialect.[2] The result is four commonly recognized Tulu dialects: North Common (NC), North Brahmin (NB), South Common (SC), and South Brahmin (SB). While Dr. Kēkuṇṇāya's work A Comparative Study of Tulu Dialects goes into detail on all differences,[3] in short, it seems that the Brahmin dialect is slightly innovative in its phonology.1

The language of the Āḷvakhēḍa (possibly OTu. āḷapa "ruling" + Skt. kheḍa "village") dynasty, which cumulatively ruled Tulunāḍṷ from the 2nd to the 14th century, is documented to have been Tulu,[4] and the poet Māmūlaṉār is known to have referenced a distinct "Tulunāṭ[u]" in the 1st century BCE-5th century CE Tamiḻ work Akanāṉūṟu.[5][6] Even considering the long history of the language, as well as its contact with written languages, until recently, it was thought that Tulu was a nonliterary language.[7] When German missionaries arrived in South Canara and transcribed the works featured in the Tübingen Tulu Manuscript, they used the Kannaḍa script,[8] due to its domination in the north of Tulunāḍṷ. Ever since, Tulu has mostly been written using a modified version of the Kannaḍa script2, and the NC dialect which was used in the manuscript is now the de facto standard version of the language.[1]

However, in 1984, the late Dr. Veṅkaṭarāja Puṇiñcattāya discovered the manuscript Śrī Bhāgavatārtho (BA),[9] which was written in a script called Tigaḷāri, a sister system of Malayāḷam Grantha which was thought to be primarily used for transcribing Sanskrit and Kannaḍa.[10] This manuscript, he believed,

[1]For example, NB, SB *pāḍanṷ* (more innovative) vs. NC, SC *pāḍandṷ* (more conservative)[3]

[2]The one clear difference between the script used for Kannaḍa and that used for Tulu is that Tulu uses the *talekaṭṭu* (ಁ) to represent the unrounded close back vowel, while Kannaḍa uses it to represent a mute consonant (i.e. with no following vowel).[1] Tulu accomodates this, since all "consonant-final" words in reality end with this vowel, and consonant clusters (*saṃyuktākṣara*) can be represented using conjuncts instead.

demonstrated a precursor to the modern form of the language: Old Tulu. This discovery led him to uncover four other "Ancient Tulu Epics": Dēvīmahātme (DM),[11] Mahābhārato (MB),[12] Kāvēri (Kv),[13] and Karṇaparvo (KP). He later published all five with additional annotations, forewords, dictionaries using the Kannaḍa script, but little work has been done to analyze the language that these books use. Dr. Kabbināle Vasanta Bharadvāja's PaḷanTulukāvya does offer some insight into its grammar, but focus is placed on classifying case suffixes into Sanskrit's eight-vibhakti system, which I don't feel does the nuances of the language enough justice, although it does provide a decent framework for the nominal system of the language.[14] The verbal system is less well understood, and therefore it is of importance in a comparative Dravidian perspective to figure out how morphologies form Tulu pieces together with other branches of the family. Also, by no means is this work comprehensive: due to the extremely limited number of copies of Dr. Puṇiñcattāya's publications in the United States, I was only able to retrieve the full texts of BA and Kv, as well as abbreviated scans of DM and MB. I hope that this description will be followed by more attempts to understand this language, possibly the missing piece to the currently unsolved puzzle of Dravidian verb morphology.

Before continuing forward, it is important to establish what the term "Old Tulu" represents. Unfortunately, all five manuscripts were written on palmleaves, which are highly perishable, and therefore, it is highly likely that the surviving ones are recopyings of the original manuscripts.[15] As a result, chemical dating is not possible, and we must resort to using events mentioned as current that can be dated using references in Kannaḍa inscriptions, as well as the saṃvatsara (Vedic year), if available. The estimates given for each work are 1200CE for DM,[11] ~1370CE for BA,[9] 1383CE for MB,[12] and 1391CE for Kv.[13] All manuscripts seem to display characteristics of the SB dialect (for example [y]ēnụ vs. NC yānụ "I" and kēmbɛ vs. NC kēṇụmbɛ "I may hear"), suggesting that the dialectal variations apparent in Modern Tulu were either present or developing in Old Tulu.[3] Therefore, this study will focus on the dialect of Tulu spoken by Brahmins in the vicinity of Kāsargōḍ between the 13th and 15th centuries

2 | A NOTE ON PHONOLOGY AND TRANSCRIPTION

Based on the writing system presented in the manuscripts, Old Tulu can be concluded to have featured the following phonemic inventory:

	Front	Central	Back	Unrounded	Rounded
Close	ī			ụ, ṵ́	ū
		i	u		
	ē				ō
Mid		a			
	e				o
	ɛ, ɛ́		ā		
Open					

Table 1: Old Tulu vowels

	L	D	A	R	P	V	G
Nasal	m	n		ṇ	ñ	ṅ	
	p	t, (ζ)		ṭ	c	k	
Stop	ph	th		ṭh	ch	kh	
	b	d		ḍ	j	g	
	bh	dh		ḍh	jh	gh	
Liquid	v	l	r	ḷ, ḻ	y		
Fricative		s, ζ		ṣ	ś		h

Table 2: Old Tulu consonants

The specific vowel qualities (specifically, the differences between long and short vowels) were chosen based on both the modern dialects of Tulu and how other Dravidian languages behave. Most consonants that are also in Sanskrit possess the same phonemic value as Sanskrit, with the exception of *r*, which is both alveolar in other Dravidian languages and is also a reflex of the Dravidian **l*, which was originally alveolar, as well. The only consonants whose pronunciations are unclear are those represented by *ḻ* and *ζ*.

ḻ represents a retroflex approximant (and in some cases, a voiced fricative) in Malayāḷam, whose script is a sister system of Tulu's. While its respective Tigaḷāri character ൰ does appear in cases where an *ḻ* appears in Malayāḷam (cf. OTu. *kaḻi-n-[a]* "which finishes" (DM 2.9)[11] and Malayāḷam *kaḻi-yuka* "to be over"),[16] it also appears as a reflex of Proto-Dravidian **r* and **ḷ* (cf. OTu. *kēḻụla* "hear!" (DM 6.2)[11] and Ta. *kēḷ* "id." as well as OTu. *baḻtenụ* "I came" (DM 1.43)[11] and Ta. *var ~ vā[r]* "to come").[17] This suggests that *ḻ* may have had allophonic representations as both a retroflex tap and a retroflex lateral approximant. I use an all-inclusive definition: retroflex liquid.

The *ζ* refers to a sound originally written as *sd* in DM and later as *sṭ* in all other texts. Its pronunciation is debated: Dr. Puṇiñcattāya posits that it is instead pronounced exactly as spelt, except that the retroflex *ṭ* becomes alveolar after the *s* (*sṯ*).[11] While this theory is superficially sound, I disagree for three reasons:

(1) Some words that are written using *sṭ* (*ζ*) are oftentimes instead spelt with *tt*: cf. OTu. *baζī* (BA 3.22.24) and [O?]Tu. *batti* "wick," and OTu. *hēpaζā* and *hēpattalā* "always."[18] These may reflect

variability in pronunciation (whether dialectal or by choice), but it is very unlikely that a single sound or group of sounds could have diverged into both *st̲* and *tt* (i.e. there is no possible candidate for reconstruction).

(2) In verse 1.188 of MB, there is a line written:

ತರಸಾ ಕೆದಿಕೀತ್‌ಸ್ಟ್ಪಣಾ ಪರಿಶಾಸ್ಟ್ ಕಳೆಲಾಕರಪೋಸ್ಟ್[12]

This would normally be transcribed as "*tarasā kedikītuζuṇā pariśāζu kaḷelākarapōζu*," but, due to the meter of this stanza (likely ˘ ˘/ ˘ ˘ / ˘ ˘/ ˘ ˘/ ˘ ˘/ ˘ ˘/\, where "\" could be either a short or long syllable[3]), it would likely be read as "*tarasā kedikītuζuṇā pariśāζ kaḷelākarapōζu*[4]," to account for the lack of a third short syllable after the fourth is. Therefore, in this example, there is a syllable that ends in a consonant cluster, which is inadmissible in Dravidian languages. As a result, *ζ* would have to either be a singular phoneme or possess an allophone pronounced with a single phoneme.

(3) Finally, descendants and cognates of words with *ζ* suggest that **st̲* was never the pronunciation.

a. First, *-ζ* is used very common as a converbial (and past?) suffix in Old Tulu, however there is no trace of either a sibilant or a voiceless retroflex in any past stem in Modern Tulu. The only two suffixes that are used to form the past in modern Tulu are *-y-* ~ *-i-* and *-t-* ~ *-d-*,[19] neither of which could possibly be linked to **st̲*. A possible theory is that *ζ* was a digraph that represented either *-y-* ~ *-i-* or *-t-* ~ *-d-*. However, it seems that the latter two were also used to form the past, leaving no reason for an extra symbol to represent these sounds. An important aspect to be considered is that Modern Tulu always uses the perfect stem (which exclusively ends in *-t-* ~ *-d-*) to form the converbial,[19] suggesting that *-ζ* may have originally been an allophonic variant of the *-t-* ~ *-d-* past.

b. Second, all forms of the distal 3rd-person singular pronoun (which can be reconstructed as **ah-t*[*ə*]) appear as modifications of *aζ-*. Notably, the nominative is *aζu̲*, accusative *aζ-e* (<**ah-t-ay*), and ablative *aζ-ertu̲* (<**ah-t ir-tt* "having been [near] that"). In Kannaḍa, the clear cognate *adu* has the variant forms *atu* and *attu*,[20] while Old Tamiḻ features a consonant-preceding form *atu* and a vowel-preceding form *ahtu*.[17] The exact phonemic value of *h* is unknown, but it is known that it was originally a separate laryngeal followed by a weak vowel, and later it oftentimes assimilated to the followed consonant and formed a geminate.[21] As a medial stage (which likely occurred in a prehistoric form of Kannaḍa), *h* would become a fricative in the following consonant's place of articulation. Tulu's use of *ζ* (a generally rare occurrence when not a converbial suffix) possibly reflects a similar pronunciation, something like **θt*.

Concludingly, I propose the following: at a stage not long before Old Tulu, **θθ* became a variant pronunciation of **tt*. As literary traditions began to materialize, **θθ* began to be used only in certain environments (i.e. converbials) leading to its complete separation from **tt*. Simultaneously, **θθ* was hypercorrected to form a variant pronunciation **θt*. **θ*, having no variant in Tigaḷāri, was transcribed as *s*, while **t* was likely transcribed as either *t* or *d*. Hence arose *sd* as the spelling for **θθ* ~ **θt*, which can be observed in DM. Why *sd/śt* became *sṭ* is enigmatic, but it's possible that, since those sound combinations are present in Sanskrit loanwords (which likely increased over time), and because *sṭ* is not generally possible in Dravidian languages (unless an archaic medial vowel is deleted), *sṭ* was used instead. At this point in time, where *ζ* preceded a vowel it would've been pronounced **θt*, whereas before a consonant it would've shortened to **θ* (< earlier **θθ*). The first pronunciation shifted to *-tt-* and was retained in all dialects but NC, while the second shifted to **ð* and then to *-d-* and was retained only in the NC.

I use *ζ* for what Tuluologists term *sṭu̲kāra* "the letter *sṭ*," since the original spelling *sd* in DM is reminiscent of the *zd* pronunciation for the letter zeta in Greek.[22] Also, since two separate spellings were used for what was likely the same phoneme, it would be much more efficient to use the same symbol in further transcription. Epsilon represents the "retroflexed *e*" vowel, while a *u* with a dot

[3]As a general rule, a syllable made up of either a short vowel or a consonant (cluster) with a short vowel would be considered a short syllable. Here, a short vowel would exclude diphthongs and anything transcribed with a macron, but it would include the *anusvāra*, written here with an *ṃ*. Syllables that have short vowels but end in single consonants can either be long or short. All other syllables are required to be long syllables.

[4]With that, I propose that the *virama*, as opposed to footnote 1, *optionally* was pronounced with the unrounded close back vowel in Old Tulu. Only recently has the vowel become required.

below represents the unrounded close back vowel. Macrons are used to denote vowel length for the vowels *a*, *e*, *i*, *o*, *u*, while an acute accent is added to show vowel length for *ɛ* and *ṵ*.[5] Finally, all citations from the epics will be italicized and in their own paragraphs, but they will not be in quotes, since they are parsed and romanized for ease of analysis.

3 | GENERAL VERB STRURE

In all Dravidian languages, verbs follow a basic strure: base-augment-termination (or stem-medial-ending). Since whatever is considered part of the base or the augment is variable in a branch's evolution, it is best to define each using benchmarks that remain in all descendants.[23] First, the base includes whatever would be included in an imperative form (excluding any imperative-specific suffixes, like . -le), meaning that, although "to think" has an imperfect stem in ninet- and a subjunctive stem in ninep-, the base is ninepu, since that is the basis of the imperative form. The termination is any suffix that indistinguishably categorizes any complete verb form but a converb (eg. ā-kụṇ- does not mean anything on its own, but ā-kụṇ-āye is recognizable as a verbal noun meaning "he who becomes"). Finally, the augment is any suffix between the base and the termination.

Tulu bases are either descended from Dravidian or borrowed from Sanskrit. Dravidian bases are derived from Proto-Dravidian roots, which are exclusively of the shape (C)VC,[24] and most suffixes that have been used to derive new bases from these roots have both lost productivity and, in many cases, nuance in meaning. Bases borrowed from Sanskrit are systematically incorporated into the Old Tulu verbal system: the suffix -(i)ta is omitted from the past passive participle and is replaced by -īppu (or its variants -īpu and -ipu). Other phonological mutation is also not uncommon; a in the first syllable is often fronted to e (eg. Skt. racita > *rac-īppu > . recīppu). For both types of bases, there are four forms of productive derivation visible in Old Tulu: causativization, effectivization, passivization, and reflexivization (see §4).

The suffixes that are part of the augment are determined by the final phoneme in the base, and, as a result, all Old Tulu verbs must be classified. As opposed to Brigel's six-class system in Modern Tulu,[19] I posit three classes for Old Tulu: I (weak); II (nasal-final); and III (strong). Strong verbs all end in the effective suffix -(p)pu, which includes all Sanskrit-derived verbs described in the previous paragraph. Nasal-final verbs are verbs that end in -ṇụ or -nụ, or possess a variant form that does so.[6] Finally, weak verbs include all other bases. The differences between these three will be further described in §5, §6, and §7, which deal with suffixes found in the augment.

The augment, at the Old Tulu stage, consists of suffixes that inflect a verb for tense, aspect, the subjunctive mood, and the optative mood. In all cases, there is exactly one suffix in the augment: suffixes for tense have an inbuilt aspeal meaning, and both the subjunctive and optative moods only apply to the nonpast. Terminations, on the other hand, include suffixes to form the conditional and imperative moods, person-number-gender suffixes, and participial suffixes.

4 | DERIVATION

The core Dravidian voice system is based on affectives and effectives, which is slightly different from the active-passive system in Indo-European. As a most basic definition, an affective verb directs an action towards the subject (i.e. the subject experiences the action, but the subject may or may not be involved in causing the action), while an effective verb directs the action towards the object. This is quite similar to an intransitive-transitive distinction, except that the verbal system is expansive enough for intransitive verbs to include passive meanings.[23] For example, "I was added by him" would be implied by the sentence "I joined because of him." Fitting with this system, there seems to have been a (set of) suffix(es) reconstructible to Proto-Dravidian that were primarily used for increasing valency (either from affective to effective, or from effective to causative), which I term "causativizers."

Primarily in South Dravidian, there is also a passive system that has developed, which usually involves the addition of an auxiliary verb meaning "to suffer." This is likely a result of contact (maybe with Sanskrit, due to its mediopassive system?), and it seems to have begun to develop after the beginning of the common era.[7] A reflexive system (implying benefit of the action to the subject) also developed (likely originally as a South Dravidian innovation, and later borrowed by Old Telugu, and then Kuvi), using the verb for "to take."[25] This might be an adaptation of Sanskrit's *ātmanepada-parasmaipada* system (cf. *paśyati* "he sees" vs. *paśyate* "he sees [for

[5] I opt for an accent mark on these vowels, instead, since the length contrast shown in manuscripts is mainly a result of meter: an *ɛ* or *ṵ* in a stressed syllable must be "lengthened."

[6] I include the second clause, since *koḷụ* ~ *koṇụ* "to take" (not to be confused with *koḷụ* "to give") and *kēḷụ* ~ *kēṇụ* "to hear" are both included in this class, even though they usually end with a lateral.

[7] I say this since Baḍaga uses the archaic affective-effective system in lieu of a more modern passive-active system, and it seems to have diverged from Old Kannaḍa just prior to its earliest written records.

himself]).[26][27] Depending on context, reflexives may also be used as passives.

Old Tulu makes use of all of these, although I separate the formation of effectives and the formation of causatives (verbs with higher valencies than effectives), since they use separate suffixes as this stage. A summary of each can be found in the table below:

Function	Suffix	Example
Effectivization	*-p(p)u*	1. *cheḍi* "come out" → *cheḍi-ppu* "cause to flow"
Causativization	*-(p(p))-ō*	1. *prāpīppu* "enter" → *prāpīp-ō* "cause to enter" 2. *paṇṵ* "speak" → *paṇṵ-p-ō* "cause to speak" 3. *ā* "become" → *āk-ō*[8] "make; cause to become"
	-(p(p))-uḍṵ	1. *nilṵ* ~ *nilpu* "stand" → *nilupuḍu* "cause to stand"
Passivization	subj. stem + *-a* + *paḍṵ*	1. *buḍṵpō* "cause to leave" → *buḍṵpō-v-a-ppaḍṵ* "be caused to leave; leave" 2. *ζṵḷepu* "start" → *ζṵḷep-a-ppaḍṵ* "be started"
Reflexivization	converb + *-oṇṵ* ~ *-oḷṵ*	1. *benṵ* "do" → *bend-oṇṵ* "do [for oneself]"
		2. *dharīppu* "assume a form" → *dharit-oḷṵ* "assume a form [for oneself]"
	subj. stem + *-oṇṵ* ~ *-oḷṵ*	1. *teḷipō* "make laugh" → *teḷipō-v-oṇṵ* "make oneself laugh" 2. *pūjipu* "worship" → *pujip-oṇṵ* "worship for oneself"

Table 3: Old Tulu verb derivation

The reflexive and passive systems, as said before, have direct parallels in other South Dravidian languages (cf. Tamiḻ *pār* "see" and *pār-ttu koḷ* "see [[for] oneself],"[17] as well as Kannaḍa *bare* "write" and *bare-yal-paḍu* "be written"[20]). The effectivizing suffix **-(p)p-* is present ubiquitously in Central Dravidian (Koṇḍa *nil-* "stand" → *nil-p-* "set up")[28] and Old Kannada (*muḍi* "finish" → *muḍipu* "cause to finish"),[29] and in trace amounts elsewhere in South Dravidian (Tamiḻ *iru* "exist, sit" → **iru-p-* > *iruvu* "seat"). The causativizing suffix is a Tulu innovation, and it likely comes from an archaic effective form of the verb *ā* "to become" (compare the *-ā* alloform of *-ō* in Modern NC Tulu). All of these forms besides the passive are present in Modern Tulu, as well.

5 | THE PAST TENSE

Whereas most other South Dravidian languages have multiple suffixes that denote the past tense, it has been long noted that Tulu possesses two pasts. Rev. J. Brigel, in *A Grammar of the Tulu Language*, and M. Rama, in *Strural description of Kasaragod Tulu*, describe the two pasts as an "imperfect" and a "[present] perfect" in the NC and SC dialects, respectively,[19][30] while Dr. Bhatt's *A Grammar of Tulu (A Dravidian Language)* and Dr. Kekunnaya's *A Comparative Study of Tulu Dialects* opt for a "recent" vs. "remote" or "past" vs. "past-perfect" distinction.[3][7] I use "imperfect" and "perfect," since they seem to be the most all-encompassing. I define "perfect" here as any description

[8]This formation is irregular, and it likely formed from an archaic intermediate **ā(k)kṵ*, found in Modern Tamiḻ as *ākku* "to make."

of a past event wherein completion (whether in the past or present) is implied, whereas "imperfect" does not imply completion.

Old Tulu seems to follow a similar system, but such nuance in meaning is difficult to discern in monolingual passages. Since there are cues, however, to the existence of the same "imperfect" vs. "perfect" in Old Tulu (such as clear ancestors of modern forms, and multiple past-tense suffixes), I attempt to apply modern classification to Old Tulu.

First, weak verbs form their imperfect by adding *-i-* ~ *-y(y)-* to the base, while the perfect is formed by adding either *-ʒ-* or *-tt-* to the base. Some examples include:

*janmādikuḷ-**āyi***[1] *trisargominī*[2] *arthādisvarajñē*[3] (BA 1.1.1)[9]
[*ā* "to become"; *ā-y-i* – become-IMPERF-PTCP]
"O Almighty, knower of[3] the three states[2] of creation, sustenance, and destruction[1]!"

asurērkuḷu[1] *dēvērkuḷe*[2] *sēneni*[3] *parājayo*[4] ***prāpipōyerụ***[5] (DM 2.4)[11]
[*prāpīpō* "cause to get"; *prāpipō-y-erụ* – cause.to.get-IMPERF-3PL.H.]
"The *Asuras*[1] caused the Gods'[2] army[3] to suffer[5] defeat[4]"

śrīmuni[1] ***cūʒī***[2] *kathe*[3] *recīppε*[4] (BA 1.19.1)[9]
[*cū* "see"; *cū-ʒ-i* – see-PERF-PTCP]
"I [may] write[4] the story[3] that the holy saint[1] has observed[2]"

kaṇṇụ[1] ***mucciʒụ***[2] *dhvānipā*[3] (BA 1.19.5)[9]
[*mucci* "close"; *mucci-ʒ* – close-PERF.cvb]
"to meditate,[3] having closed[2] the eyes[1]"

mōhontāva[1] ***kūḍụttī***[2] *manassụḷḷāye*[3] (DM 1.22)[11]
[*kūḍụ* "join"; *kūḍụ-tt-i* – join-PERF-PTCP]
"a man with a heart[3] filled[2] with love[1]"

aḷav[1]-***ātụn***[2]*-ī*[3] *yuga*[4]*-vartako*[5] (BA 1.1.28)[9]
[*ā* "become"; *ātụnụ* – become-PERF-3SG.N.]
"the state[5] of this[3] era[4] has become[2] strong[1]"

Nasal-final verbs form their perfect and one imperfect through use of a secondary stem, but in rare cases, the imperfect is formed by *-i-* ~ *-y(y)-* to the the bare base. The secondary stem is formed by adding either a *-d-* to a final *-n(ụ)* or a *-ḍ-* to a final *-ṇ(ụ)*. Finally, the more common imperfect is equivalent to the secondary stem, while adding *-tt-* to the secondary stem forms the perfect.

kēḷina*-ndo* (BA 1.1.28)[9]
[*kēḷụ* ~ *kēṇụ* "hear"; *kēḷ-i-na* – hear-IMPERF-PTCP]
"like hearing"

*vēda-vyāsere vandit-**oṇḍe*** (BA 1.2.3)[9]
[*koḷụ* ~ *koṇụ* "take"; *koṇ-ḍ-e* – take-IMPERF-3SG.M.]
"the knowing *Vyāsa* saluted"

*kauṣṭikerụ vijñāpano **vendi** vacano **kēṇḍụttụ** mārkaṇḍēyerụ prativacano ājñāpīpperụ* (DM 1.5)[11]
[*venụ* "do"; *ven-d-i* – do-IMPERF-PTCP; *kēṇụ* "hear"; *kēṇ-ḍ-tt* – hear-PERF.cvb]
"having heard the words that *Kauṣṭika* spoke, *Mārkaṇḍēya* tells [his] response"

*parājayo prāpiti dēverkuḷ-āʒụnākuḷu tāmarettụ-ṇḍānañcittī brahmadēvereni numb-**ijivoṇḍụttụ*** (DM 2.6)[11]
[*ijivoḷụ* ~ *ijivoṇụ* "place"; *ijivoṇ-ḍ-tt* – place-PERF.cvb]
"the gods who were defeated, having previously placed *Brahma*, borne out of a lotus"

Finally, strong verbs also form their perfect and imperfect with a secondary stem, which is formed by replacing the final *-(p)pu* with a *-tụ*. While it is not a hard-and-fast rule, it is also very common for *-īppu* verbs, which oftentimes are Sanskrit loanwords, to become *-itụ* (rather than *-ītụ*, for example). Just like nasal-final verbs, the imperfect is equivalent to the secondary stem, and adding *-tt-* to the secondary stem forms the perfect.

kēṇḍ[1]*-anēko-la*[2] ***mōdit***[3]*-eṅkuḷu*[4] (Kv 6.105)[13]
[*mōdīppu* "be happy"; *mōdi-t* – be.happy-IMPERF.cvb]
"we[4], who have been joyed[3] by all[2] that [we] heard[1]"

cūyina[1] *sādhu*[2]*-saṅgom*[3]*-**apēkṣitụ***[4] (Kv. 6.110)[13]
[*apēkṣīppu* "desire"; *apēkṣi-t* – desire-IMPERF.cvb]
"having wished for good company, which [he?] saw"

jātivēṣomin[1]*-**vañcitụttina***[2] *bhinnomiṃ*[3] *buḍụ*[4] *akṣiṇaṃ*[5] (Kv. 6.114)[13]
[*vañcīppu* "conceal"; *vañci-t-tt-ina* – conceal-PERF-PTCP]
"Leave[4], at once[5], the other [form?][3] which had concealed[2] the human-form[1]"

kaṅkaṇī[1] ***kụḍatụtti***[2] (BA 1.1.12)[9]
[*kụḍāppu* "lie down"; *kụḍa-t-tt-i* – lie.down-PERF-PTCP]
"…which had lain down[2] pronely[1]"

Thus, it can be concluded that the perfect can be formed for all verbs using the suffix *-tt-* or its alloform *-ʒ-* (see §2), and the imperfect can either be formed by adding *-i-* ~ *-(y)y-* (where a secondary stem is not needed) or *-Ø-* (where a secondary stem is required).

There are two possibilities regarding origins of the Old Tulu past tense:

(1) First, considering that one past stem of the verb for "to be" is *ṷtt-*, it is possible that the perfect stems of the verbs in Class II and Class III are aally periphrastic constructions with this *ṷtt-* past stem. As a result, it could be posited that, for Class II and Class III, the secondary stems (or the imperfect stem) were the original "past-tense forms," while, in Class I, *-tt-* (<3 morae) or *-i(y)-* (3+ morae) would be added to form the past depending on base strure. Later, the periphrastic perfect in Class II and Class III became widespread, leading to a reworking of the Class I past system, where **-tt-* and **-i(y)-* were both generalized with slightly different meanings as past suffixes for all Class I verbs: **-tt-*, analogizing to *-ṷtt-* in the other two classes, became the perfect, and **-i(y)-* became the imperfect.

a. This is tempting, considering the verbal systems in other South Dravidian languages. For example, in Tamiḻ and Kannaḍa, verbs that are less than three morae exclusively take a dental past suffix of the forms *-t-*, *-nt-*, or *-tt-* (Tamiḻ) and *-d-*, *-nd-*, or *-t-* ~ *-tt-* (Kannaḍa), whereas verbs that are three or more morae long take *-i* in both languages.[24] Also, this creates striking correspondences between past stems in Tamiḻ-Kannaḍa, and "original" past stems in Pre-Old Tulu:

Proto-South-Dravidian	Modern Tamiḻ	Old Kannaḍa	Pre-Old Tulu
**tin* [DEDR 3263]	*tin-ṟ-* < **tin-t-*	*tin-d-*	**tin-d-*
**toṭ-ā-nkᵘ* [DEDR 3481]	*tuṭaṅk-i-*	*toḍag-i-*	**tuḍeṅg-i-*
**aẓ* [DEDR 282]	*aḻu-t-*	*aḻ-t-*	**aḻu-tt-*
**aṇ(ṇ)-ā*[9] [DEDR 110]	*aṇṇā-nt-*	-	**aṇṇā-tt-*

Table 4: Cognate Past Stems between South Dravidian languages[31]

b. However, the suggestion that two separate suffixes, *-i-* and *-T-*,[10] originally had the same meaning, but were specific to verbs of different base shape, is questionable. Not only that, but there are hints to more free variation between *-i-* and *-T-* pasts in the earliest Tamiḻ and Kannaḍa texts, suggesting that the Old Tulu description of the verb is ancient.[11]

c. At the same time, there was no driving force for *-tt-* to have spread to verbs of more than three morae in Old Tulu, as suggested by this theory. Tulu could have as well created a perfect for these verbs by adding their converb in *-i* to *-ṷtt-*. This would've resulted in **tuḍeṅg-i-ṷtt-en*, for example, for "he has started," and later **tuḍeṅg-itt-en*. There would've been no areal pressure either, since no other South Dravidian language adds *-T-* pasts to trimoraic verbs. Therefore, this rule is likely an innovation in the Tamiḻ-Kannaḍa subgroup.

d. For these reasons, I suggest a second theory:

(2) As Krishnamurti has stated in *The Dravidian Languages*, Dravidian roots are always monosyllabic and of the shapes (C)V(C) and (C)V̄(C).[24] As a result, any attested verbs that are longer than this are formed with the help of derivational suffixes or as lexicalized converbs[12] (where the augment is incorporated into the base). At the same time, Krishnamurti has also noted that **-p(p)-* and **-t(t)-* are attested as effectivizing/causativizing suffixes all across Dravidian.[24] Therefore, I posit that the imperfect-perfect distinction was original (possibly with slightly different labels), and the imperfect and perfect stems in Class I are direct descendants of Proto-South Dravidian, with slight change and generalization with the dental suffix. Class III verbs each originally had two interchangeable bases, derived from the **-p(p)-* and **-t(t)-* suffixes. These later became confused with the homophonous nonpast and past suffixes *-pp-* and *-tt-*, and each pair of verbs collapsed into one base with a pair of stems, where the causative in **-t(t)-*

[9]This does not fit into the three-mora rule, since almost all trimoraic verbs that are not modern compounds or borrowings have a final epenthetic vowel (*-ᵘ* in Tamiḻ and *-u* in Kannaḍa). This instead is part of Class III in Old Tulu.

[10]I use this symbol to account for all the different dental pasts in Tamiḻ and Kannaḍa.

[11]Rajam, in *A Reference Grammar of Old Tamil Poetry*, states that all classes of Old Tamiḻ verbs can form a past in *-i-* or *-y-*, although trimoraic verbs cannot take a *-T-* past.[32] In Old Kannaḍa, there are some cases where verbs have a "short" converb, some of which display a *-T-* past for trimoraic verbs: *aḍigu* "to hide" ⟶ *aḍik* "having hidden" < **aṭik-(t)t-* << **aṭ-i-k-* [DEDR 83].[20][31]

[12]As an example, the verb *tiruku* "twist" in Tamiḻ is formed from the root **tir* [DEDR 3246][31] and the optative-imperfective tense suffix *-k-*.

formed the secondary base (for the perfect and imperfect), while the causative in **-p(p)-* formed the primary base (for the nonpast and irrealis moods). For Class II verbs, the original perfect stem (**-d-* ~ **-ḍ-*) formed a lexicalized converb, which was later repurposed as the secondary stem. Each secondary stem in Class II and III had originally taken the same perfect and imperfect suffixes (**-tt-* and **-i(y)-*), but the *-y-* in the imperfect stem later elided.

a. There are many reasons to suggest that this was the origin. Two derived bases of the same root collapsing into different stems of the same base is not uncommon. In fact, the same process likely occurred with causatives in Old Tamiḻ: *iruv-um* (< **ir-p-un* ~ **ir-p-ump^u*) "it causes to sit" vs. *irīi* (**ir-c-i-*) "having caused to sit, arising from the **-p(p)-* and **-c(c)-* causative suffixes.
b. Lexicalized converbs of this form are uncommon in the Tamiḻ-Kannaḍa subgroup of South Dravidian, but there are attested examples of similar formations in Old Tulu. For example, the verb *nendṵ* is derived from the verb *nilpu* (< **nil-t-*), although there are isolated forms where *nendṵ* acts as a converb of *nilpu*, such as in *nendṵ-pu-le* (stand.cvb exist-IMP). Therefore, this might be considered another example, where a lexicalized converb is used (probably at the Pre-Old stage) as the secondary stem, whereas *nilpu* is used as the the primary stem.
c. Elision of the *-y-* suffix is aally quite common, especially in Central Dravidian. For example, in the coastal dialects of Telugu, *cēs-in-ānu* ~ **cēs-i-ānu* "I did" becomes *cēśæːnu*.[33] A parallel process, *-v-* elision, which is described further in §7, can also be observed. The subjunctive of all verbs originally was *-v-*, but it later became *-Ø-* for most. It is only retained in verbs that end in a vowel: *tīruv-āpaḷē* "at the time of ending" (Kv 6:112)[13]; *tīru-v-a* – end-SUBJ-PTCP. Finally, there is evidence that an archaic imperfect stem was later used to create the infinitive of purpose (see §10) for some verbs, such as in *paṇ-iy-era* "in order to speak."

6 | THE NONPAST TENSE

For all classes of verbs, the nonpast is formed by adding the suffix *-pp-*, which is sometimes shortened to *-p-*. For Classes II and III, this suffix is added to the primary stem. See below:

bēṇṭekṵ[1] ***pōpp[ɛ]***[2] (DM 1.17)[11]
[*pō* "go"; *pō-pp-ɛ* – go-NPST-1SG]
"I will go[2] on a hunt[1]"

mahiṣāsura[1] *bādhettāva*[2] ***saṅkaṭopaḍṵppi-ñcittī***[3] *dēvērkuḷāζṵnakuḷu*[4] (DM 2.36)[11]
[*saṅkaṭopaḍṵ* "suffer"; *saṅkaṭopaḍṵ-pp-i* – suffer-NPST-PTCP]
"the gods,[4] who suffer[3] the torment[2] of Mahiṣāsura[1]"

paṇṵppuṇ[1]*-eṅkuḷu*[2] (Kv 6.110)[13]
[*paṇṵ* "say, speak"; *paṇṵ-pp-uṇa* – say-NPST-PTCP]
"we[2], who say[1]"

ēṇalōcane[1] *garbhomeñca*[2] *dharītero*[3]*-ndṵ*[4] ***pinṵppirī***[5] (BA 1.1.33)[9]
[*pinṵ* "know"; *pinṵ-pp-iri* – know-NPST-NEG]
"I do not know[5] whether[4] the deer-eyed woman[1] became[3] pregnant[2]"

parātpara[1]*-mūrti*[2] *vanditṵ*[3] ***varṇipuppɛn***[4]*-atī*[5] *kathe*[6] (BA 1.1.5)[9]
[*varṇīppu* "describe"; *varṇīppu-pp-ɛnṵ* – describe-NPST-1SG]
"saluting[3] god's[1] statue[2], I will describe[4] the story[6] in detail[5]"

ambujapriya[1]*-numbuṭṵ*[2] ***nalipuppi***[3] *menpulit*[4]*-andome*[5] (BA 1.1.14)[9]
[*nalipu* "dance"; *nalipu-pp-i* – dance-NPST-PTCP]
"like[5] a firefly[4], which dances[3] in[2] the sun[1]"

The *-pp-* suffix directly corresponds with the *-v-*, *-p-*, *-b-*, and *-pp-* found in the rest of South Dravidian. This is a recent innovation, which was likely areally motivated (due to the nearby Kannaḍa and Malayāḷam), and it was likely originally a Class III subjunctive suffix along with *-mb-* (Class II) and *-v-* (Class I). It replaced earlier **-n-*, which is discussed further in §10.

7 | THE SUBJUNCTIVE AND OPTATIVE MOODS

The subjunctive and optative suffixes in Dravidian have always been grouped with the nonpast and have been treated as tense suffixes, since they usually signify future events. A trifold distinction between realis, subjunctive, and optative has mostly disintegrated in modern Dravidian

languages, although a singular impersonal optative (which makes use of the reconstructed optative suffix **-k(k)-*) is usually retained in South Dravidian languages.[13] Kuvi and Baḍaga are peculiar in retaining the irrealis meaning of **-p(p)-*,[14] but this seems to cover the optative, as well, allowing **-k(k)-* to drop out of usage.

Old Tulu seems to be the only South Dravidian language to retain complete optative and subjunctive paradigms that are distinct in meaning, although they are sometimes used in place of nonpasts. Modern Tulu possesses the same subjunctive system, but the Old optative is now limited to the first person,[3] similar to nearby Dravidian languages.

With regards to the subjunctive, Class II verbs lose the final nasal and take the suffix *-mb-*:

pimbane (BA 1.1.14)[9]
[*pinụ* "know"; *pi^n-mb-ane* – know-SUBJ-NEG.PTCP]
"without knowing"

puṇyādhikērkuḷū-nduñca[1] ***paṇụpōvomberụ***[2] (DM 4.50)[11]
[*paṇụpōvoṇụ* "be called"; *paṇụpōvo^n-mb-erụ* – be.called-SUBJ-3PL.H]
"[they] may be called[2] meritorious people[1]"

All other verbs take *-Ø-* if they are consonant-final (*-ụ*) or *-v-* if they are vowel-final. Alternatively, a euphonic *-u* may be added to a consonant-final upon the writer's discretion, on which case *-v-* is used.

sambhramippɛn[1]*-atī*[2] *kathē*[3] *terit*[4]*-ompe*[5] *varṇiperā-nduṭụ*[6] (BA 1.14)[9]
[*sambhramīppu* "explain"; *sambhramīpp-Ø-ɛnụ* – explain-SUBJ-1SG]
"I am excited[1] to explain[6] fully[2] the story[3] as far[5] as [I] know[5]"

bandhuvargoṅkụḷekụḷā[1] *dukkho*[2] ***ζụṇḍāvanụ***[3] (DM 4.49)[11]
[*ζụṇḍā* "arise, come into existence"; *ζụṇḍā-v-anụ* – arise-SUBJ-NEG.3SG.N.]
"Grief[2] may not come[3] upon the group of relatives[1]"

ζorme[1] ***sēruvom***[2] (Kv 6.112)[13]
[*sērụ* "join"; *sēr-uv-om* – join-SUBJ-1PL]
"we may join[2] [as] one[1]"

Optatives for all classes are formed by adding the suffix *-k-* to the (primary) base:

tani[1] ***pinki***[2] *prakāro*[3] (BA 1.1.1)[9]
[*pinụ* "know"; *pin-k-i* – know-OPT-PTCP]
"the way[3] that [he] knows[2] himself[1]"

nālụ[1] *diśeṭ-uḷḷa*[2] *jeno*[3] ***mānipukụṇa****-ndo*[4] (Kv 7.2)[13]
[*mānipu* "respect"; *mānipu-k-uṇa* – respect-OPT-PTCP]
"the way that the people of all four directions respect [him]"

cūker (MB 1.188)[12]
[*cū* "see"; *cū-k-erụ* a – see-OPT-PTCP]
"they see"

8 | SUBJECT-VERB AGREEMENT

Suffixes can be added to form both finite verbs and verbal nouns. While finite verbs can be formed for all subjects, verbal nouns can only be formed for third-person subjects and second-person plural subjects. Verbal noun terminations can only be added to participles, not bare verb stems, whereas finite verb terminations must be added to base verb stems.

	1	2	3, masc.	3, fem.	3, neu.
singular	*-ɛ(nụ)*	*-a*	*-e(nụ)*	*-aḷụ*	*-nụ ~ -ṇụ ~ -ṇu, -u*
plural	*-o(m)*	*-arụ*	*-erụ*		*-o*

Table 5: Personal terminations in finite verbs

	1	2	3, masc.	3, fem.	3, neu.
singular			*-āye*	*-āḷụ*	*-ai, -aye, -aζụ*
plural		*-ārụ*	*-ākuḷu*		*-ai*

Table 6: Personal terminations in verbal nouns

9 | NEGATION

There are three main routes of forming negatives in Old Tulu – (1) adding *-(u)ddi* to a participle, (2) adding *-iri* to a tense-mood stem, and (3) adding *-a(n)-* ~ *-av-* ~ *-ay-* to the subjunctive stem to form a negative subjunctive. Both are likely descendants of **inṟ^u* "it does not exist" (< **il-t* [DEDR 2559]),[31] each of which developed divergently. The second route is similar to Old Tamiḻ, wherein *-il-* is added between a converb and a personal termination (eg. *cey-t-*

[13]Literary Tamiḻ *-(k)ka*,[34] Kannaḍa *-ge* ~ *-ke*,[29] Koḍagu "negative potential" *-vaku āga*[35]

[14]Kuvi *nānu wā-p-ee* "I may come"[36] and Baḍaga *nānu bembe* "let me/may I bake [i.e. in the sun]." I thank Anandhan Raju for the Baḍaga example and translation.

il-eṉ "I do/did not do"), and the *-ā* negative converbial suffix (eg. *ceyy-ā* "without doing").[37] Examples include:

yena[1] *putrērkuḷe*[2] *naḍavaḷā*[3] ***pinụpṇaddi***[4] (DM 1.44)[11]
[*pinụ* "know"; *pinụ-pp-uṇa-ᵘddi* – know-NPST-PTCP-NEG]
"I don't know[4] [of] the behavior[3] of my[1] sons[2]"

niṣṭhurākōppi[1] *manassụ*[2] ***puṭṭụpụṇaddi***[3] (DM 1.50)[11]
[*puṭṭụ* "be born"; *puṭṭụ-pp-uṇa-ᵘddi* – be.born-NPST-PTCP-NEG]
"a heart[2] that causes enmity[1] is not born[3]"

rañjipottiri (BA 1.13)[9]
[*rañjīppō* "please"; *rañjīppō-tt-iri* – please-PERF-NEG]
"[it] didn't please"

vēdajñerụ[1] *śāstromi*[2] *sūkṣitụlā*[3] ***manipuppiri***[4] (BA 1.19)[9]
[*manipu* "speak"; *manipu-pp-iri* – speak-NPST-NEG]
"those who know the Vedas, having studied them, do not speak"

paramārtho[1] ***cūvayer***[2] (BA 1.17)[9]
[*cū* "see"; *cū-v-ay-erụ* – see-SUBJ-NEG-3PL.H]
"they may not see[2] the highest truth[1]"

harisatkathe[1]-***āvane***[2] (BA 1.18)[9]
[*ā* "be, become"; *ā-v-an-e* – be-SUBJ-NEG-cvb.]
"without being Vishnu's story"

10 | NON-FINITE FORMS

There are five categories for non-finite verbs in Old Tulu – imperative, conditional, infinitive, converb, and participle. The imperative is formed by adding the suffix *-le* to the primary base, while the conditional is constructed by adding *-uṭa* to the subjunctive base of a verb. The infinitive is formed by adding *-era* to Class III verbs, and adding *-(i)yera* to Class II and Class I verbs.

-le should be compared to *-ali* in Medieval and Modern Kannaḍa,[29] both of which likely come from the ancient verbal noun in **-al*. There is another suffix *-elụ* in Modern Tulu,[18] used to form nouns from verbs, which is clearly a descendant of this same suffix, meaning that *-le* likely arose as a borrowing from Kannaḍa. *-uṭa* is *-ṇṭa* in Modern Tulu for the SC and SB dialects,[3] as well as closely related Mūḍu Koraga,[38] and they can be added directly to finite verbs, suggesting that they may have originally been from an auxiliary verb. I propose that this is archaic conditional form of the verb "to exist," created similar to Telugu and Tamiḻ conditionals: **uḷ-tt-āl* (be-PST-COND) > **uṇ-ṭṭ-ā* ~ **uṇ-ṭṭ-a* > *-ṇṭa*; cf. Old Telugu *camp-in-an* (kill-PST-COND), Modern Telugu *aṇ-ṭ-ē* (say-PST-COND),[33] and Literary Tamiḻ *cey-t-āl* (do-PST-COND).[34]

Generally, auxiliary verbs are usually added to the subjunctive or imperfect stem of a verb in Old Tulu (and lose their initial consonant), for example *vandit-oḷụ* < **vanditụ koḷụ* "revere [for oneself], be revered" (imperfect stem), *khaṇḍitụḍụyerụ* < **khaṇḍitụ buḍụyerụ* "they cut into pieces[15]" (imperfect stem), and *stutīppōvoṇụppuṇārụ* < **stutīppō-v-a koṇụppuṇārụ* "they who praise themselves" (subjunctive stem). It is possible that, first, the *-era* comes from an archaic verb meaning "to exist,"[16] and the *-i* is an ancient imperfect stem, which is not attested in Class II verbs in Old Tulu, even though the *-(i)yera* form applies to both it and Class I. The *-era* form in Class III is probably the same auxiliary verb appended to a subjunctive stem.

yeṅkụ[1] *vistārontāva*[2] ***ājñāpipul[e]***[3] (DM 1.4)[11]
[*ājñāpīppu* "tell"; *ājñāpīppu-le* – tell-IMP]
"tell [it][3] to me[1] in detail[2]"

sūkṣit[1]-***ōrppile***[2] (BA 1.2.7)[9]
[*ōrppu* "listen"; *ōrppu-le* – listen-IMP]
"diligently[1] listen[2]"

īśvarārpaṇomāva[1] ***bembuṭa***[2] (Kv 6.107)[13]
[*benụ* "do, work"; *be*ⁿ*-mb-uṭa* – do-SUBJ-COND]
"if [one] does [it][2] as an offering to God[1]"

kīrtitā[1] *śile*[2]-*tīrthom*[3]-*ēnya*[4] *buḍante*[5] *sēviter*[6]-***āvuṭa***[7] (Kv 6.111)[13]
[*ā* "be, become"; *ā-v-uṭa* – be-SUBJ-COND]
"if[7] they serve[6], without ever[4] dropping[5] a bit of[2] water[3], out of praise[1]"

īyyụ[1]... *īḍekāva*[2] ***baryera***[3] *jāẓụ*[4] *kāraṇo*[5] (DM 1.38)[11]
[*barụ* "come"; *bar-iyera* – come-INF]
"what[4] is the reason[5] for you[1] to come[3] here[2]"

nityo[1] ***śōbhipera***[2] (BA 1.2.13)[9]
[*śōbhīppu* "shine"; *śōbhīpp-era* – shine-INF]
"in order to shine[2] every day[1]"

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[15]Possibly a completive. Compare cognate *kaṇṭittu-viṭu* in Literary Tamiḻ.[17]

[16]Compare Tamiḻ-Kannaḍa *iru*[17][20] and Koraga unreal past *-ern-*,[38] which is likely a nonpast stem of **ir* "exist" [DEDR 480].[31] At the same time, the final *-a* is a common infinitive suffix in all South Dravidian languages aside from Modern Malayāḷam,[24] meaning *-era* could be the infinitive form of this verb in Pre-Old Tulu. Also note that the passive is formed with a suffix *-a*, as is in Tamiḻ,[34] where it is considered to be derived from the infinitive suffix.

Since sentences in most Dravidian languages can only contain a single independent clause, converbs simulate independent clauses (without a subject), while also showing relation in time to the main verb. Positive converbs are equivalent in form to perfect and imperfect stems, whereas negative converbs are formed by adding any of the suffixes *-ane*, *-ante*, *-antettu* or *-anettu*. The negative converb suffixes *-ane* and *-ante* may be considered to be tense suffixes attached to the negative suffix *-a* (< Proto-South Dravidian **-ah-*); this process is also characteristic of Tamiḻ (cf. *ceyy-ā-mē*,[39] which features both a negative and nonpast suffix) and Kannaḍa (cf. *hēḷ-a-de*,[29] which is composed of a negative and a past suffix). *-anettu* and *antettu* could be better analyzed as *-ane-ettu* and *-ante-ettu*, and therefore is composed of the negative converb and the positive perfect converb of an archaic verb meaning "to be" described in the section of infinitives (*ettu* < **er-ttu*).

kṣaṇo[1] *portuṭṭu*[2] *bāṇominin*[3] ***buḍuζú***[4] (MB 1.192)[12]
[*buḍu* "leave, let go"; *buḍu-ζ* – leave-PERF.cvb]
"having discharged[4] an arrow[3] in the span[2] of an instant[1]"

saṅkulitu[1] *vartipoḍu*[2] *sādhukuḷu*[3] ***pindu***[4] (Kv 7.67)[13]
[*saṅkulipu* "be delighted," *saṅkuli-tu* – be.delighted-IMPERF.cvb; *pinu* "know," *pin-du* – know-IMPERF.cvb]
"saints,[3] having known [that],[4] should act[2] delightedly[1]"

śrīkathē[1] ***viracīpanē***[2] (BA 1.2.11)[9]
[*viracīppu* "compose"; *viracīp-ane* – compose-NEG.cvb]
"without composing[2] a grand story[1]"

kīrtitā[1] *śile*[2]*-tīrthom*[3]*-ēnya*[4] ***buḍante***[5] *sēviter*[6]*-āvuṭa*[7] (Kv 6.111)[13]
[*buḍu* "leave, let go"; *buḍ-ante* – leave-NEG.cvb]
"if[7] they serve[6], without ever[4] dropping[5] a bit of[2] water[3], out of praise[1]"

rūpu[1] ***cōjovanettu***[2] (BA 1.1.5)[9]
[*cōjō* "cause to appear, create"; *cōjō-(v)-anettu* – create-NEG.cvb]
"without creating[2] a form[1]"

ele[1] *suratha*[2] *cakrēśvare*,[3] *dēvērkuḷāsduṇākukuḷu*[4] *ākuḷeṭa*[5] *yuddho*[6] ***benantett***[7]*-ē*[8] *pidaḍyerु*[9] (DM 5.6)[11]
[*benu* "do, work"; *ben-antettu* – do-NEG.cvb]
"O[1] King[3] Suratha,[2] the gods[4] left[9] without even[8] fighting[7] a war[6] with them![5]"

Participles are used in lieu of relative clauses, and two suffixes can be used to form them. *-i*, *-na*, or *-ina* can be added directly to all tense/mood stems, whereas *-uṇa* is restricted to the nonpast and optative. Negative participles are created by adding the aforementioned suffixes to *-ante* and *-an(t)ettu*, forming *-antina*, *-antettina*, and *-anettuṇa*, for example. *-i* might be related to the third-person masculine singular termination *-i* in Koraga, as well as the Old Tamiḻ masculine agentive suffix *-i* (cf. *ēnt-i* "holder") found in bhakti texts.[37] *-uṇa* is likely the verb for "to be," *uḷ*, with the Proto-South Dravidian nonpast suffix **-n-* and the participial suffix *-a* suffixed afterward.

Nonpast participles can also be formed by adding *-n-* directly to the (primary) verbal base. This is derived from the aforementioned reconstructed nonpast **-n-*, attested in Tamiḻ in nonpast verbal nouns like *varu-n-ar* "they who come."

buddhiṭu[1] ***sthāpitī***[2] *sacarācaraṃ*[3] (BA 1.1.5)[9]
[*sthāpīppu* "create"; *sthāpi-t-i* – create-IMPERF-PTCP]
"a universe[3] created[2] from [his] mind[1]"

nigamō[1]*-**nkina***[2] *kalpatarun*[3]*-tudiṭú*[4] *phalom*[5]*-āζ*[6]*-osaguppī*[7] (BA 1.1.3)[9]
[*enu* "say"; *en-k-ina* – say-OPT-PTCP]
"…which appears[7] as[6] a fruit[5] at the tip[4] of a boon-granting tree[3] called[2] *nigama* (the Vedas)[1]"

putrērkuḷeni[1] *pitāvu*[2] *yeñca*-***rakṣipukuṇa***[3] *ā*[4] *prakāromē*[5] (DM 1.13)[11]
[*rakṣīppu* "protect"; *rakṣīppu-k-uṇa* – protect-OPT-PTCP]
"that[4] way[5] which a father[2] protects[3] [his] sons[1]"

maṅkaṇī[1] ***teriyanti***[2] *mūḍhe*[3] *samudro*[4] *nīndukuṇa*[5]*-ndome*[6] (BA 1.1.12)[9]
[*teri* "be known, be clear"; *teri-(y)-ant-i* – be.known-NEG-PTCP]
"like[6] a fool[3] who doesn't know[2] how to keep [his] head up[1] swimming[5] the oceans[4]"

tatvomē ***pinanetti*** *jentukuṇ… tatvomī* ***pinanettuṇ-ākḷekuṇ*** (Kv 6.111) [13]
[*pinu* "know"; *pin-anett-i* – know-NEG-PTCP, *pin-anett-uṇa* – know-NEG-PTCP]
"for the animals that don't understand reality… for the [people] that don't understand reality "

urasthaloṇṭuttu[1] ***uṇḍāna***-*ñcittī*[2] *putrērkuḷeni*[3] (DM 1.13)[11]
[*uṇḍā* "come into existence, be born"; *uṇḍā-na* – be.born-NPST.PTCP]
"sons[3] borne[2] from the chest[1]"

11 | CONCLUSION

This paper intended to provide a basis for further research into Old Tulu verb morphology, as verbs are the

lesser understood category of words within Dravidian. Although this description is not as extensive, it is hoped that future works can update claims posited here and paint a much more detailed pire of Tulu. More surveying must be done to isolate the nuanced meaning of each suffix, and possibly discover previously unseen verb morphology. Not only that, the subtle morphophonemic processes that occur extensively in these texts need to be understood better. Finally, more attention must be paid to syntax and serial verb construction, since these are present, not just in Tulu, but virtually everywhere in Dravidian. Old Tulu, as a newly discovered language with a large corpus, creates the opportunity for a review of our current understanding of historical and comparative Dravidian linguistics.

S1 | SAMPLE PARADIGMS

(1) Class I: cū "see"

	Perfect1	Perfect2	Imperfect
1sg	cūʒɛ(nụ)	cūttɛ(nụ)	cūyɛ(nụ)
1pl	cūʒo(m)	cūtto(m)	cūyo(m)
2sg	cūʒa	cūtta	cūya
2pl	cūʒarụ	cūttarụ	cūyarụ
3sgm	cūʒe(nụ)	cūtte(nụ)	cūye(nụ)
3sgf	cūʒaḷụ	cūttaḷụ	cūyaḷụ
3sgn	cūʒụṇụ	cūttụṇụ	cūyụṇụ ~ cūyinụ
3plh	cūʒerụ	cūtterụ	cūyerụ
3pln	cūʒo	cūtto	cūyo

	Nonpast	Subjunctive	Optative
1sg	cūppɛ(nụ)	cūvɛ(nụ)	cūkɛ(nụ)
1pl	cūppo(m)	cūvo(m)	cūko(m)
2sg	cūppa	cūva	cūka
2pl	cūpparụ	cūvarụ	cūkarụ
3sgm	cūppe(nụ)	cūve(nụ)	cūke(nụ)
3sgf	cūppaḷụ	cūvaḷụ	cūkaḷụ
3sgn	cūppụṇụ	cūvu	cūkụṇụ?
3plh	cūpperụ	cūverụ	cūkerụ
3pln	cūppo	cūvo	cūko

(2) Class II: enụ "say"

	Perfect	Imperfect
1sg	endɛ(nụ)	endụttɛ(nụ)
1pl	endo(m)	endụtto(m)
2sg	enda	endụtta
2pl	endarụ	endụttarụ
3sgm	ende(nụ)	endụtte(nụ)
3sgf	endaḷụ	endụttaḷụ
3sgn	endụṇụ	endụttụṇụ
3plh	enderụ	endụtterụ
3pln	endo	endụtto

	Nonpast	Subjunctive	Optative
1sg	enụppɛ(nụ)	embɛ(nụ)	enụkɛ(nụ)
1pl	enụppo(m)	embo(m)	enụko(m)
2sg	enụppa	emba	enụka
2pl	enụpparụ	embarụ	enụkarụ
3sgm	enụppe(nụ)	embe(nụ)	enụke(nụ)
3sgf	enụppaḷụ	embaḷụ	enụkaḷụ
3sgn	enụppụṇụ	embu	enụkụṇụ?
3plh	enụpperụ	emberụ	enụkerụ
3pln	enụppo	embo	enụko

(3) Class III: kụḍāppu "lie down"

	Perfect	Imperfect
1sg	kụḍatɛ(nụ)	kụḍatụttɛ(nụ)
1pl	kụḍato(m)	kụḍatụtto(m)
2sg	kụḍata	kụḍatụtta
2pl	kụḍatarụ	kụḍatụttarụ
3sgm	kụḍate(nụ)	kụḍatụtte(nụ)
3sgf	kụḍataḷụ	kụḍatụttaḷụ
3sgn	kụḍatụṇụ	kụḍatụttụṇụ
3plh	kụḍaterụ	kụḍatụtterụ
3pln	kụḍato	kụḍatụtto

	Nonpast	Subjunctive	Optative
1sg	kụḍāppuppɛ(nụ)	kụḍāppɛ(nụ)	kụḍāppukɛ(nụ)
1pl	kụḍāppuppo(m)	kụḍāppo(m)	kụḍāppuko(m)
2sg	kụḍāppuppa	kụḍāppa	kụḍāppuka
2pl	kụḍāppupparụ	kụḍāpparụ	kụḍāppukarụ
3sgm	kụḍāppuppe(nụ)	kụḍāppe(nụ)	kụḍāppuke(nụ)
3sgf	kụḍāppuppaḷụ	kụḍāppaḷụ	kụḍāppukaḷụ
3sgn	kụḍāppuppụṇụ	kụḍāppu	kụḍāppukụṇụ?
3plh	kụḍāppupperụ	kụḍāpperụ	kụḍāppukerụ
3pln	kụḍāppuppo	kụḍāppo	kụḍāppuko

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